

Chapter-5

Coordinate systems

Cartesian coordinates:

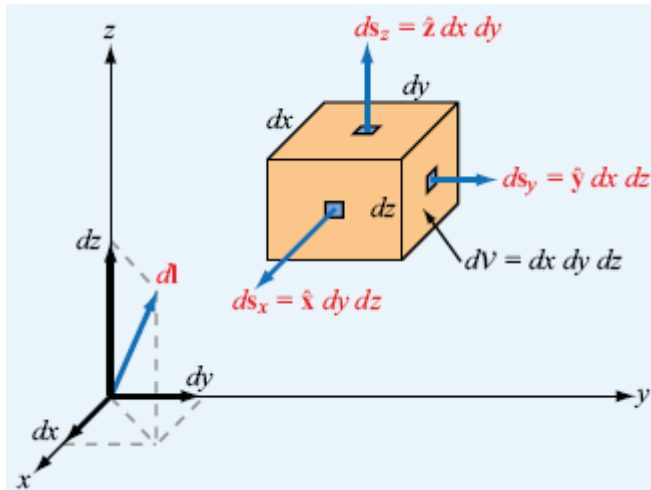


Figure: Differential length, area, and volume in Cartesian coordinates.

Cartesian variables: x, y, z

Vector representation: $\vec{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$

Magnitude of $\vec{A}$: $|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$

Position vector: $\vec{OP} = \hat{x}x_1 + \hat{y}y_1 + \hat{z}z_1$ for $P = (x_1, y_1, z_1)$

Basic vector properties: $\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = \hat{z} \cdot \hat{z} = 1$ and $\hat{x} \cdot \hat{y} = \hat{y} \cdot \hat{x} = \hat{z} \cdot \hat{x} = 0$

$\hat{x} \times \hat{y} = \hat{z}, \hat{y} \times \hat{z} = \hat{x}, \hat{z} \times \hat{x} = \hat{y}$

Dot product: $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

Cross product: $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{x}(A_y B_z - A_z B_y) - \hat{y}(A_x B_z - A_z B_x) + \hat{z}(A_x B_y - A_y B_x)$

Differential length: $d\vec{l} = \hat{x} dx + \hat{y} dy + \hat{z} dz$

Differential surface area: $d\vec{s}_x = \hat{x} dy dz, d\vec{s}_y = \hat{y} dz dx, d\vec{s}_z = \hat{z} dx dy$

Differential volume: $dV = dx dy dz$

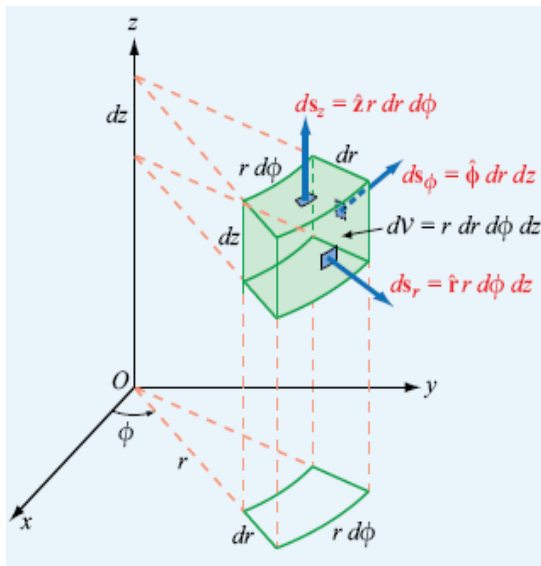
Cylindrical coordinates:

Figure: Differential length, area, and volume in cylindrical coordinates.

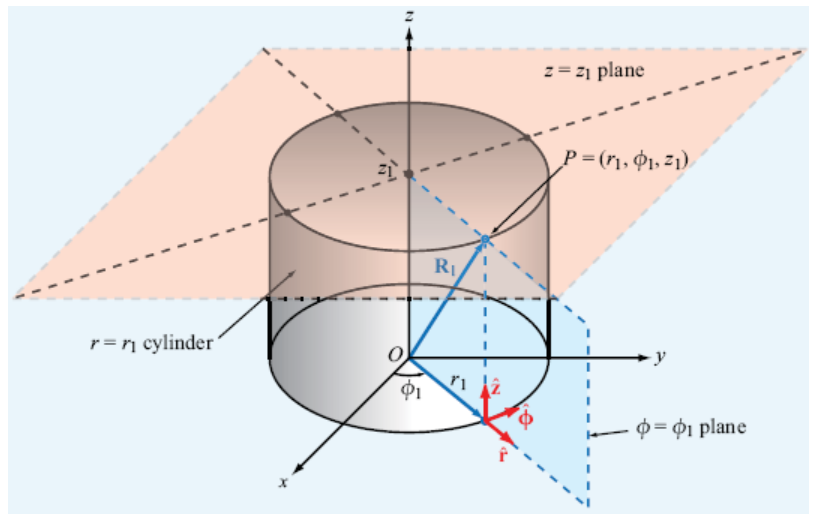


Figure: Point $P(r_1, \phi_1, z_1)$ in cylindrical coordinates; r_1 is the radial distance from the origin in the xy plane, ϕ_1 is the angle in xy plane measured from the x axis toward the y axis, and z_1 is the vertical distance from the xy plane.

Cylindrical variables: r, ϕ, z

Vector representation: $\vec{A} = \hat{r}A_r + \hat{\phi}A_\phi + \hat{z}A_z$

Magnitude of $\vec{A}$: $|\vec{A}| = \sqrt{A_r^2 + A_\phi^2 + A_z^2}$

Position vector: $\vec{OP} = \hat{r}r_1 + \hat{\phi}\phi_1 + \hat{z}z_1$ for $P = (r_1, \phi_1, z_1)$

Basic vector properties: $\hat{r} \cdot \hat{r} = \hat{\phi} \cdot \hat{\phi} = \hat{z} \cdot \hat{z} = 1$ and $\hat{r} \cdot \hat{\phi} = \hat{\phi} \cdot \hat{z} = \hat{z} \cdot \hat{r} = 0$

$$\hat{r} \times \hat{\phi} = \hat{z}, \hat{\phi} \times \hat{z} = \hat{r}, \hat{z} \times \hat{r} = \hat{\phi}$$

Dot product: $\vec{A} \cdot \vec{B} = A_r B_r + A_\phi B_\phi + A_z B_z$

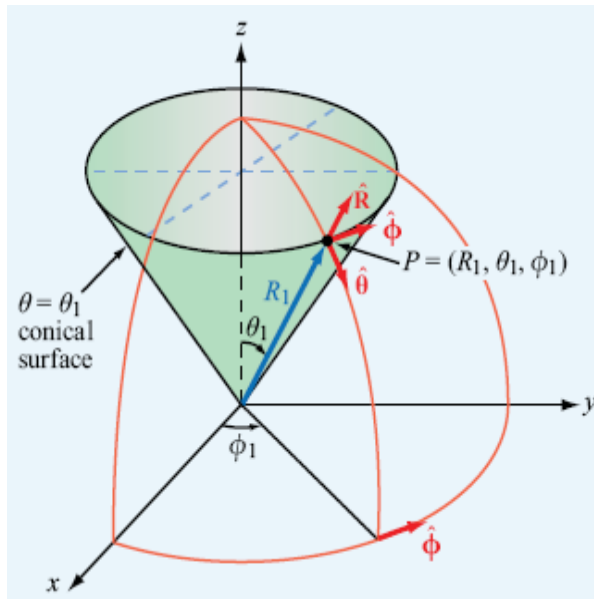
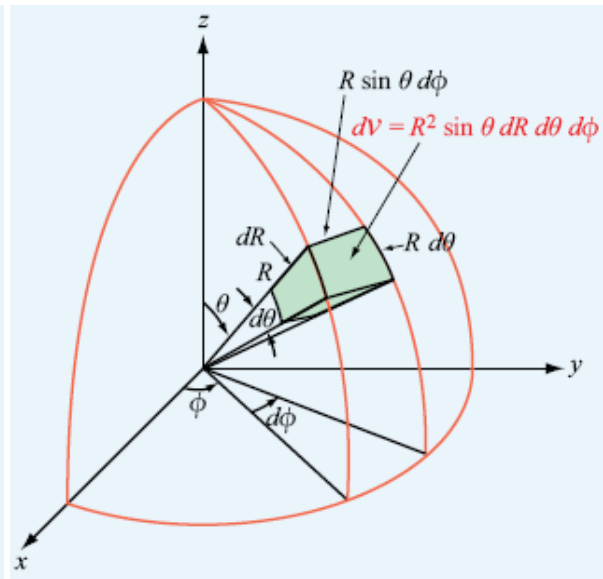
Cross product:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{r} & \hat{\phi} & \hat{z} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix} = \hat{r}(A_\phi B_z - A_z B_\phi) - \hat{\phi}(A_r B_z - A_z B_r) + \hat{z}(A_r B_\phi - A_\phi B_r)$$

Differential length: $d\vec{l} = \hat{r}dr + \hat{\phi}r d\phi + \hat{z}dz$

Differential surface area: $d\vec{s}_r = \hat{r}r d\phi dz$, $d\vec{s}_\phi = \hat{\phi}dr dz$, $d\vec{s}_z = \hat{z}dr r d\phi$

Differential volume: $dV = dr r d\phi dz$

Spherical coordinates:**Figure:** point $P(R_1, \theta_1, \phi_1)$ in spherical coordinates.**Figure:** Differential volume in spherical coordinates.Spherical variables: R, θ, ϕ Vector representation: $\vec{A} = \hat{R}A_r + \hat{\theta}A_\theta + \hat{\phi}A_\phi$ Magnitude of $\vec{A}$: $|\vec{A}| = \sqrt{A_r^2 + A_\theta^2 + A_\phi^2}$ Position vector: $\vec{OP} = \hat{R}R_1 + \hat{\theta}\theta_1 + \hat{\phi}\phi_1$ for $P = (R_1, \theta_1, \phi_1)$ Basic vector properties: $\hat{R} \cdot \hat{R} = \hat{\theta} \cdot \hat{\theta} = \hat{\phi} \cdot \hat{\phi} = 1$ and $\hat{R} \cdot \hat{\theta} = \hat{\theta} \cdot \hat{\phi} = \hat{\phi} \cdot \hat{R} = 0$ $\hat{R} \times \hat{\theta} = \hat{\phi}$, $\hat{\theta} \times \hat{\phi} = \hat{R}$, $\hat{\phi} \times \hat{R} = \hat{\theta}$ Dot product: $\vec{A} \cdot \vec{B} = A_r B_r + A_\theta B_\theta + A_\phi B_\phi$

Cross product:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{R} & \hat{\theta} & \hat{\phi} \\ A_r & A_\theta & A_\phi \\ B_r & B_\theta & B_\phi \end{vmatrix} = \hat{R}(A_\theta B_\phi - A_\phi B_\theta) - \hat{\theta}(A_r B_\phi - A_\phi B_r) + \hat{\phi}(A_r B_\theta - A_\theta B_r)$$

Differential length: $d\vec{l} = \hat{R} dR + \hat{\theta} R d\theta + \hat{\phi} R \sin \theta d\phi$ Differential surface area: $d\vec{s}_R = \hat{R} R d\theta R \sin \theta d\phi$, $d\vec{s}_\theta = \hat{\theta} dR R \sin \theta d\phi$,

$$d\vec{s}_\phi = \hat{\phi} dR R d\theta$$

Differential volume: $dV = dR R d\theta R \sin \theta d\phi$

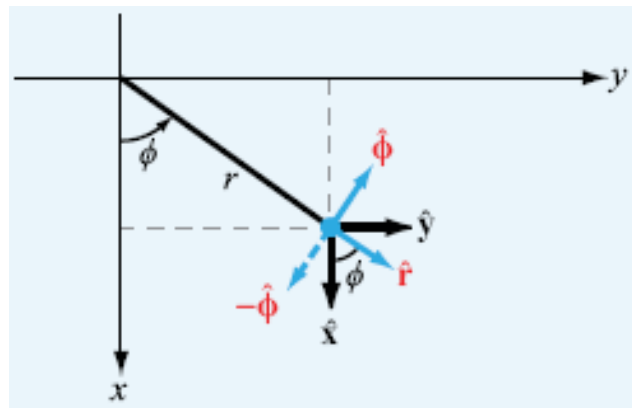
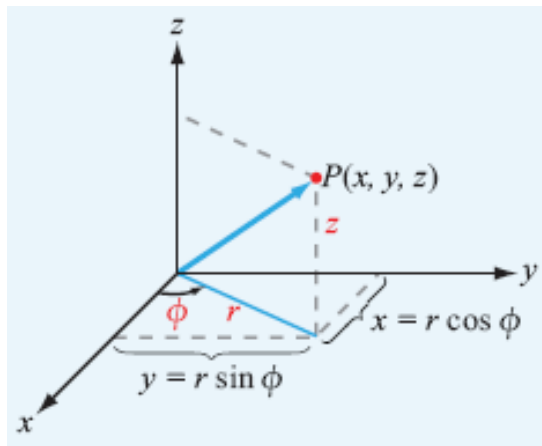
Transformation (Cartesian coordinates to cylindrical and vice versa):

Figure: Interrelationships between Cartesian coordinates (x, y, z) and cylindrical coordinates (r, ϕ, z) . **Figure:** Interrelationships between base vectors $(\hat{x}, \hat{y})$ and $(\hat{r}, \hat{\phi})$.

We know, $x = r \cos \phi$, $y = r \sin \phi$ and $z = z$

$$r = \sqrt{x^2 + y^2}, \phi = \arctan\left(\frac{y}{x}\right), z = z; \quad 0 \leq r < \infty, \quad 0 \leq \phi < 2\pi, \quad -\infty < z < \infty$$

		A_r	A_ϕ	A_z
		$\hat{r}$	$\hat{\phi}$	$\hat{z}$
A_x	$\hat{x}$	$\cos \phi$	$-\sin \phi$	0
A_y	$\hat{y}$	$\sin \phi$	$\cos \phi$	0
A_z	$\hat{z}$	0	0	1

That is, $\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi$, $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$ and $\hat{z} = \hat{z}$.

$$A_r = A_x \cos \phi + A_y \sin \phi, \quad A_\phi = -A_x \sin \phi + A_y \cos \phi \quad \text{and} \quad A_z = A_z.$$

$$\hat{x} = \hat{r} \cos \phi - \hat{\phi} \sin \phi, \quad \hat{y} = \hat{r} \sin \phi + \hat{\phi} \cos \phi, \quad \hat{z} = \hat{z}$$

$$A_x = A_r \cos \phi - A_\phi \sin \phi, \quad A_y = A_r \sin \phi + A_\phi \cos \phi, \quad A_z = A_z$$

Transformation (Spherical coordinates to cylindrical coordinates and vice versa):

We know, $r = R \sin \theta$, $z = R \cos \theta$ and $\phi = \phi$

$$R = \sqrt{r^2 + z^2}, \theta = \arctan\left(\frac{r}{z}\right), \phi = \phi; \quad 0 \leq r < \infty, \quad 0 \leq \phi < 2\pi, \quad -\infty < z < \infty$$

		A_R	A_θ	A_ϕ
		$\hat{R}$	$\hat{\theta}$	$\hat{\phi}$
A_r	$\hat{r}$	$\sin \theta$	$\cos \theta$	0
A_z	$\hat{z}$	$\cos \theta$	$-\sin \theta$	0
A_ϕ	$\hat{\phi}$	0	0	1

That is, $\hat{r} = \hat{R} \sin \theta + \hat{\theta} \cos \theta$, $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$ and $\hat{\phi} = \hat{\phi}$.

$$A_r = A_R \sin \theta + A_\theta \cos \theta, \quad A_z = A_R \cos \theta - A_\theta \sin \theta \quad \text{and} \quad A_\phi = A_\phi.$$

$$\hat{R} = \hat{r} \sin \theta - \hat{z} \cos \theta, \quad \hat{\theta} = \hat{r} \cos \theta - \hat{z} \sin \theta, \quad \hat{\phi} = \phi$$

$$A_R = A_r \sin \theta - A_z \cos \theta, \quad A_\theta = A_r \cos \theta - A_z \sin \theta, \quad A_\phi = A_\phi$$

Transformation (spherical coordinates to Cartesian coordinates and vice versa):

We know, $x = R \sin \theta \cos \phi$, $y = R \sin \theta \sin \phi$ and $z = R \cos \theta$;

$$R = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \arctan\left(\frac{\sqrt{x^2 + y^2}}{z}\right), \quad \phi = \arctan\left(\frac{y}{x}\right);$$

$$0 \leq r < \infty, \quad 0 \leq \phi < 2\pi, \quad -\infty < z < \infty$$

		A_R	A_θ	A_ϕ
		$\hat{R}$	$\hat{\theta}$	$\hat{\phi}$
A_x	$\hat{x}$	$\sin \theta \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
A_y	$\hat{y}$	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$\cos \phi$
A_z	$\hat{z}$	$\cos \theta$	$-\sin \theta$	0

That is,

$$\hat{x} = \hat{R} \sin \theta \cos \phi + \hat{\theta} \cos \theta \cos \phi - \hat{\phi} \sin \phi, \quad \hat{y} = \hat{R} \sin \theta \sin \phi + \hat{\theta} \cos \theta \sin \phi + \hat{\phi} \cos \phi$$

$$\text{and } \hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta.$$

$$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi,$$

$$A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$$

$$\text{and } A_z = A_R \cos \theta - A_\theta \sin \theta.$$

$$\hat{R} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta$$

$$\hat{\theta} = \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta$$

$$\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$$

$$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$$

$$A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

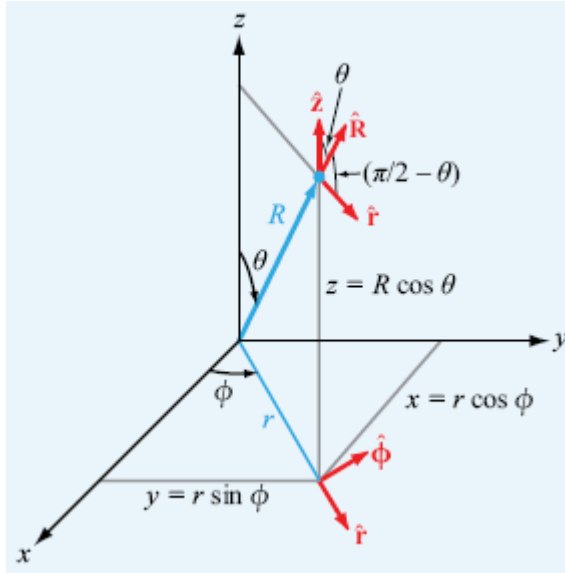


Figure: Interrelationships between (x, y, z) and (R, θ, ϕ)

Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$ $\hat{z} = \hat{z}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{x} = \hat{r} \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{r} \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{z}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{R} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta$ $\hat{\theta} = \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{x} = \hat{R} \sin \theta \cos \phi + \hat{\theta} \cos \theta \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{R} \sin \theta \sin \phi + \hat{\theta} \cos \theta \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{R} = \hat{r} \sin \theta + \hat{z} \cos \theta$ $\hat{\theta} = \hat{r} \cos \theta - \hat{z} \sin \theta$ $\hat{\phi} = \hat{\phi}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{r} = \hat{R} \sin \theta + \hat{\theta} \cos \theta$ $\hat{\phi} = \hat{\phi}$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$

Example 1: Transform $\left(\sqrt{2}, \frac{3\pi}{4}, 3\right)$ from cylindrical coordinates to Cartesian coordinates.

Solution: $x = r \cos \phi = \sqrt{2} \cos \frac{3\pi}{4} = -1,$

$$y = r \sin \phi = \sqrt{2} \sin \frac{3\pi}{4} = 1,$$

$$z = z = 3$$

So, Cartesian point $(x, y, z) = (-1, 1, 3)$

Example 2: Transform $(-1, 1, 3)$ from Cartesian coordinate to cylindrical coordinate.

Solution: $r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$

$$\phi = \tan^{-1}\left(\frac{1}{-1}\right) = \tan^{-1}(-1) = \frac{3\pi}{4}$$

$$z = z = 3$$

Example 3: Transform $(1, 0, \sqrt{3})$ from Cartesian coordinate to spherical coordinate.

Solution: $R = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + 0^2 + (\sqrt{3})^2} = \sqrt{4} = 2$

$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right) = \tan^{-1}\left(\frac{\sqrt{1^2 + 0^2}}{(\sqrt{3})^2}\right) = \frac{\pi}{6}$$

$$\phi = \tan^{-1}\frac{y}{x} = \tan^{-1}\frac{0}{1} = 0$$

So, the spherical point $(R, \theta, \phi) = \left(2, \frac{\pi}{6}, 0\right)$

Example 4: Transform $\left(2, \frac{\pi}{6}, 0\right)$ from spherical coordinate to Cartesian coordinate.

Solution: $x = R \sin \theta \cos \phi = 2 \sin \frac{\pi}{6} \cos 0 = 1$

$$y = R \sin \theta \sin \phi = 2 \sin \frac{\pi}{6} \sin 0 = 0$$

$$z = R \cos \theta = 2 \cos \frac{\pi}{6} = \sqrt{3}$$

So, the Cartesian point $(x, y, z) = (1, 0, \sqrt{3})$

Example 5: Transform $\left(2, \frac{\pi}{6}, 0\right)$ from spherical coordinate to cylindrical coordinate.

Solution: $r = R \sin \theta = 2 \sin \frac{\pi}{6} = 1$

$$z = R \cos \theta = 2 \cos \frac{\pi}{6} = \sqrt{3}$$

$$\phi = \phi = 0$$

So, the cylindrical point $(r, z, \phi) = (1, \sqrt{3}, 0)$

Example 6: Transform $(1, 0, \sqrt{3})$ from cylindrical coordinate to spherical coordinate.

Solution: $R = \sqrt{r^2 + z^2} = \sqrt{1^2 + (\sqrt{3})^2} = 2$

$$\theta = \tan^{-1}\left(\frac{r}{z}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\phi = \phi = 0$$

So, the spherical point $(R, \theta, \phi) = \left(2, \frac{\pi}{6}, 0\right)$

Example 7: Express vector $\vec{A} = \hat{x}(x + y) + \hat{y}(y - x) + \hat{z}z$ in spherical coordinate.

Solution: We know in spherical coordinate

$$\vec{A} = \hat{R} A_R + \hat{\theta} A_\theta + \hat{\phi} A_\phi$$

$$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$$

$$= (x + y) \sin \theta \cos \phi + (y - x) \sin \theta \sin \phi + z \cos \theta$$

$$= (R \sin \theta \cos \phi + R \sin \theta \sin \phi) \sin \theta \cos \phi$$

$$+ (R \sin \theta \sin \phi - R \sin \theta \cos \phi) \sin \theta \sin \phi + R \cos \theta \cos \theta$$

$$= R \sin^2 \theta \cos^2 \phi + R \sin^2 \theta \sin \phi \cos \phi + R \sin^2 \theta \sin^2 \phi - R \sin^2 \theta \sin \phi \cos \phi$$

$$+ R \cos^2 \theta$$

$$= R \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + R \cos^2 \theta$$

$$= R \sin^2 \theta + R \cos^2 \theta = R$$

$$A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$$

$$= (x + y) \cos \theta \cos \phi + (y - x) \cos \theta \sin \phi - z \sin \theta$$

$$= (R \sin \theta \cos \phi + R \sin \theta \sin \phi) \cos \theta \cos \phi$$

$$+ (R \sin \theta \sin \phi - R \sin \theta \cos \phi) \cos \theta \sin \phi - R \sin \theta \cos \theta$$

$$= R \sin \theta \cos \theta \cos^2 \phi + R \sin \theta \cos \theta \sin \phi \cos \phi + R \sin \theta \cos \theta \sin^2 \phi$$

$$- R \sin \theta \cos \theta \sin \phi \cos \phi - R \sin \theta \cos \theta$$

$$= R \sin \theta \cos \theta (\sin^2 \phi + \cos^2 \phi) - R \sin \theta \cos \theta$$

$$= 0$$

$$\begin{aligned}
A_\phi &= -A_x \sin \phi + A_y \cos \phi \\
&= -(x + y) \sin \phi + (y - x) \cos \phi \\
&= -(R \sin \theta \cos \phi + R \sin \theta \sin \phi) \sin \phi + (R \sin \theta \sin \phi - R \sin \theta \cos \phi) \cos \phi \\
&= -R \sin \theta \sin \phi \cos \phi - R \sin \theta \sin^2 \phi + R \sin \theta \sin \phi \cos \phi - R \sin \theta \cos^2 \phi \\
&= -R \sin \theta (\sin^2 \phi + \cos^2 \phi) = -R \sin \theta
\end{aligned}$$

$$\therefore \vec{A} = \hat{R} R - \hat{\phi} R \sin \theta$$

Example 8: Express vector $\vec{A} = \hat{R} R - \hat{\phi} R \sin \theta$ in Cartesian form.

Solution: We know in Cartesian coordinate $\vec{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$

$$\begin{aligned}
A_x &= A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi \\
&= R \sin \theta \cos \phi + 0 \cdot \cos \theta \cos \phi + R \sin \theta \sin \phi \\
&= x + y
\end{aligned}$$

$$\begin{aligned}
A_y &= A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi \\
&= R \sin \theta \sin \phi + 0 \cdot \cos \theta \sin \phi - R \sin \theta \cos \phi \\
&= y - x
\end{aligned}$$

$$\begin{aligned}
A_z &= A_R \cos \theta - A_\theta \sin \theta \\
&= R \cos \theta - 0 \cdot \sin \theta \\
&= z
\end{aligned}$$

$$\therefore \vec{A} = \hat{x}(x + y) + \hat{y}(y - x) + \hat{z}z$$

Example 9: Transform vector $\vec{A} = \hat{x}(x + y) + \hat{y}(y - x) + \hat{z}z$ to cylindrical coordinates.

Solution: We know, cylindrical coordinate $\vec{A} = \hat{r}A_r + \hat{\phi}A_\phi + \hat{z}A_z$

$$\begin{aligned}
A_r &= A_x \cos \phi + A_y \sin \phi = (x + y) \cos \phi + (y - x) \sin \phi \\
&= (r \cos \phi + r \sin \phi) \cos \phi + (r \sin \phi - r \cos \phi) \sin \phi \\
&= r \cos^2 \phi + r \sin \phi \cos \phi + r \sin^2 \phi - r \sin \phi \cos \phi \\
&= r(\cos^2 \phi + \sin^2 \phi) = r
\end{aligned}$$

$$\begin{aligned}
A_\phi &= -A_x \sin \phi + A_y \cos \phi \\
&= -(x + y) \sin \phi + (y - x) \cos \phi \\
&= -(r \cos \phi + r \sin \phi) \sin \phi + (r \sin \phi - r \cos \phi) \cos \phi \\
&= -r \cos \phi \sin \phi - r \sin^2 \phi + r \cos \phi \sin \phi - r \cos^2 \phi \\
&= -r(\sin^2 \phi + \cos^2 \phi) \\
&= -r
\end{aligned}$$

$$A_z = z$$

$$\therefore \vec{A} = \hat{r}r - \hat{\phi}r + \hat{z}z$$

Example 10: Express vector $\vec{A} = \hat{r} r - \hat{\phi} r + \hat{z} z$ in Cartesian coordinate.

Solution: We know in Cartesian coordinate $\vec{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$

$$A_x = A_r \cos \varphi - A_\varphi \sin \varphi$$

$$= r \cos \varphi + r \sin \varphi$$

$$= x + y$$

$$A_y = A_r \sin \varphi + A_\varphi \cos \varphi$$

$$= r \sin \varphi - r \cos \varphi$$

$$= y - x$$

$$A_z = z$$

$$\text{Hence, } \therefore \vec{A} = \hat{x}(x + y) + \hat{y}(y - x) + \hat{z}z$$

Example 11: Transform the vector $\vec{A} = \hat{r}3 \cos \phi - \hat{\phi}2r + \hat{z}z$ to spherical coordinate.

Solution: We know in spherical coordinate $\vec{A} = \hat{R}A_R + \hat{\theta}A_\theta + \hat{\phi}A_\phi$

$$A_R = A_r \sin \theta + A_z \cos \theta$$

$$= 3 \cos \phi \sin \theta + z \cos \theta = 3 \cos \phi \sin \theta + R \cos \theta \cos \theta$$

$$A_\theta = A_r \cos \theta - A_z \sin \theta$$

$$= 3 \cos \phi \cos \theta - z \sin \theta = 3 \cos \phi \cos \theta - R \cos \theta \sin \theta$$

$$A_\phi = A_\phi = -2r = -2R \sin \theta$$

$$\therefore \vec{A} = \hat{R}(3 \cos \phi \sin \theta + R \cos^2 \theta) - \hat{\theta}2R \sin \theta + \hat{\phi}(3 \cos \phi \cos \theta - R \cos \theta \sin \theta)$$

Example 12: Transform the vector $\hat{A} = \hat{R} \cos \phi + \hat{\theta} \sin \phi + \hat{\phi} \sin^2 \theta$ at the point $\left(3, \frac{\pi}{2}, \pi\right)$ to cylindrical coordinates.

Solution: Given that, $\hat{A} = \hat{R} \cos \phi + \hat{\theta} \sin \phi + \hat{\phi} \sin^2 \theta$ at the point $\left(3, \frac{\pi}{2}, \pi\right)$.

Here, $R = 3$, $\theta = \frac{\pi}{2}$, $\phi = \pi$ then $r = R \sin \theta = 3$, $\phi = \pi$ and $z = R \cos \theta = 0$

We get, $\left(3, \frac{\pi}{2}, \pi\right) \rightarrow (3, \pi, 0)$

Now, from given equation $A_R = \cos \phi = -1$, $A_\theta = \sin \phi = 0$, $A_\phi = \sin^2 \theta = 1$

We know,

$$\hat{A} = A_r \hat{r} + A_\phi \hat{\phi} + A_z \hat{z}$$

$$\therefore A_r = A_R \sin \theta + A_\theta \cos \theta = -1, A_\phi = A_\phi = 1 \text{ and } A_z = A_R \cos \theta - A_\theta \sin \theta = 0$$

Hence, $\hat{A} = -\hat{r} + \hat{\phi}$.

Example 13: Transform the vector $\vec{A} = \hat{r} \cos \phi - \hat{\phi} \sin \phi + \hat{z} \sin \phi \cos \phi$ at the point $\left(2, \frac{\pi}{4}, 2\right)$ to spherical coordinates.

Solution: Given that, $\vec{A} = \hat{r} \cos \phi - \hat{\phi} \sin \phi + \hat{z} \sin \phi \cos \phi$ at the point $\left(2, \frac{\pi}{4}, 2\right)$.

Here, $A_r = \cos \phi$, $A_\phi = -\sin \phi$, and $A_z = \sin \phi \cos \phi$ and $r=2$, $\phi=\frac{\pi}{4}$, $z=2$

We get, $R = \sqrt{r^2 + z^2} = \sqrt{4 + 4} = \sqrt{8}$, $\theta = \tan^{-1}\left(\frac{r}{z}\right) = \tan^{-1}\left(\frac{2}{2}\right) = \frac{\pi}{4}$, $\phi = \frac{\pi}{4}$

$$\therefore (R, \theta, \phi) = \left(\sqrt{8}, \frac{\pi}{4}, \frac{\pi}{4}\right)$$

We know, $\vec{A} = A_R \hat{R} + A_\theta \hat{\theta} + A_\phi \hat{\phi}$

Here, $A_R = A_r \sin \theta + A_z \cos \theta$, $A_\theta = A_r \cos \theta - A_z \sin \theta$ and $A_\phi = A_\phi$.

$A_R = \cos \phi \sin \theta + \sin \phi \cos \phi \cos \theta$, $A_\theta = \cos \phi \cos \theta - \sin \phi \sin \phi \sin \theta$ and $A_\phi = -\sin \phi$.

$$\therefore \vec{A} = (\cos \phi \sin \theta + \sin \phi \cos \phi \cos \theta) \hat{R} + (\cos \phi \cos \theta - \sin \phi \sin \phi \sin \theta) \hat{\theta} + (-\sin \phi) \hat{\phi}$$

$$\text{At the point } \left(\sqrt{8}, \frac{\pi}{4}, \frac{\pi}{4}\right) \quad \vec{A} = .853 \hat{R} + 0.1464 \hat{\theta} - \frac{1}{\sqrt{2}} \hat{\phi}.$$

Sample exercise-5.1

1. Convert the coordinates of the following points from Cartesian to cylindrical and spherical coordinates:

- a. $p_1 = (-\sqrt{2}, \sqrt{2}, 2)$
- b. $p_2 = (0, 2, 2\sqrt{3})$
- c. $p_3 = (4\sqrt{3}, 4, -8)$

2. Convert the coordinates of the following points from cylindrical to Cartesian and spherical coordinates:

- a. $p_1 = (3, \frac{5\pi}{3}, 3)$
- b. $p_2 = (4, \frac{\pi}{4}, 0)$
- c. $p_3 = (2\sqrt{3}, \frac{\pi}{6}, -2)$

3. Convert the coordinates of the following points from spherical to Cartesian and cylindrical coordinates:

- a. $p_1 = (\sqrt{2}, \frac{3\pi}{4}, \pi)$
- b. $p_2 = (2, \frac{\pi}{4}, \frac{3\pi}{4})$
- c. $p_3 = (1, \frac{\pi}{4}, \frac{\pi}{2})$

4. Transform the following vectors into cylindrical coordinates at the indicated points:

- a. $A = \hat{x}(x+y) + \hat{y}(x+y) + \hat{z}z$ at $p = (4, 0, -4)$ **Ans:** $A = 4\hat{r} + 4\hat{\phi} - 4\hat{z}$.
- b. $B = \hat{R} \sin \theta + \hat{\theta} \cos \theta + \hat{\phi} \cos^2 \phi$ at $p = (2, \frac{\pi}{2}, \frac{\pi}{4})$ **Ans:** $B = \hat{r} + \frac{1}{2}\hat{\phi}$.

5. Transform the following vectors into spherical coordinates at the indicated points:

- a. $A = \hat{x}y - \hat{y}x$ at $p = (1, -1, 0)$ **Ans:** $A = -\sqrt{2}\hat{\phi}$.
- b. $B = \hat{z} \sin \phi$ at $p = (2, \frac{\pi}{4}, 2)$ **Ans:** $B = \frac{1}{2}\hat{R} - \frac{1}{2}\hat{\theta}$.

6. Transform the following vectors into cartesian coordinates at the indicated points:

- a. $A = -\hat{\phi}R \sin \theta$ at $p = (1, \frac{\pi}{2}, 0)$ **Ans:** $A = -\hat{y}$.
- b. $B = -\hat{r} \cos \phi - \hat{\phi} \sin \phi + \hat{z}z$ at $p = (2, \frac{2\pi}{3}, 2\sqrt{3})$ **Ans:** $B = \frac{1}{2}\hat{x} + \frac{\sqrt{3}}{2}\hat{y} + 2\sqrt{3}\hat{z}$.